PalmSource Developer Conference 2004
FEBRUARY 10-12, 2004 SAN JOSE, CA



Presented By:
JB Parrett
Principal Software Developer

POWERED

Going Native: Porting to ARM

Tips, Tricks and Traps

The contents of and the technologies described in this presentation are provided to you and described "AS IS," without any warranty or representation of any kind. PalmSource hereby disclaims each and every other representation or warranty with respect thereto, including the implied warranties of merchantability, fitness for a particular purpose, accuracy and non-infringement. No representation or warranties are made that any product based in whole or in part on any specifications or information referenced herein will be free from defects or safe for use for its intended purposes.

PalmSource further reserves the right to modify in any respect any technology described or content provided herein without prior written notice.

Palm OS is a registered trademark and Palm Powered is a trademark of Palm Trademark Holding Company, LLC. Other brands may be trademarks of their respective owners. All other trademarks and tradenames used herein are the property of their respective owners.

Copyright (c) 2004 PalmSource, Inc.



Overview

What Do You Mean, “Porting?”

How to Prepare

Things Your Compiler Warned You About…

…And Things It Can’t

Q & A



Why You Can't Just Recompile

We're not in Kansas anymore…

- Processor
- Application runtime environment
- APIs
- Shared libraries
- PIM databases

Goals:

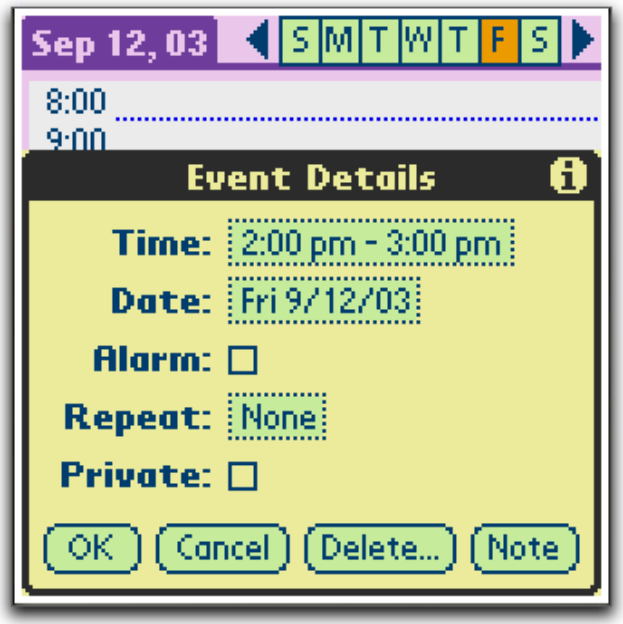
- Minimize source changes
- Reduce initial time to port
- Fix/improve APIs
- Support 'old ways' (for a while)

Preparing to Port Your Application

Test your current 68K application

- Emulator with Palm OS® 4 Debug ROM
- Palm OS® 5 Simulator
- Palm OS® Cobalt® Simulator
- **Debug** ROMs, and *all* debug checks enabled

Fix errors *before* porting





What Could Possibly Go Wrong?

Does your code:

- Use System-only APIs?
- Use undocumented resources?
- Directly access hardware or display memory?
- Access UI data structure?
- Access low memory globals?
- Handle frmUpdateEvent properly?
- Run with scissors? *

* Running with scissors to be supported in next release



Porting Steps

1. Read *Porting Applications to Palm OS Cobalt*
2. Convert app resources to XRD format
3. Move sources and resources to new IDE
4. Compile!
5. Fix compiler errors
6. Test!
7. Fix non-compiler errors



Resources

Clear the Save Behind flag for modal dialogs

- Before or after resource conversion

GenerateXRD and GenerateXRD Wizard

Designed to accept .rsrc files, PRCs, XRDs

- Windows CW: use file in Resource.frk directory
- Mac OS CW: convert to Windows format first
- PilRC: create a .ro file or a .prc

Designed to inline bitmaps and other binary data

- Or will save in a directory and reference in XRD



What Will I Have to Change?

Trivial

- Changes that can be deferred
- Changes that can be automated

Simple

- Minor adjustments
- Changes with side effects

Less simple, more typing

- Function rewrite

Logic adjustment

- Thinking and typing required

Compile-Time

Run-Time



Header Files

Some are gone…

- `PalmOSGlue.h`
- `PalmUtils.h`
- `PalmOSCompatibility.h`

Some are new…

- `CatMgr.h`
- `EventCodes.h`
- `ErrTryCatch.h`

Use `PalmOS.h` in most cases



Changes You Can Defer

Many API changes are covered by compatibility header files

- Function name or parameter list differences

Examples:

Old Name	New Name
`UInt16`	`uint16_t`
`DmFindSortPosition()`	`DmGetRecordSortPosition()`
`DmCreateDatabase()`	`DmCreateDatabaseV50()`
`SortRecordInfoType`	`DmSortRecordInfoType`



Changes You Can Automate

Fix with global search and replace

- Or your own compatibility macros

Required when no compatibility header exists

- Or use of compatibility header not desired

Designed to work if data type size is same, or if parameter can be easily matched in search:

- Find: `FrmGotoForm(formID)`
- Replace with: `FrmGotoForm(dbRef, formID)`



Minor Adjustments

`ROMVersionCompatible()`

- Check for 6.0 (Release) as minimum version
- Palm OS Cobalt apps will not launch on older devices
- Remove check for Palm OS 1.0

Get app's resource `DmOpenRef` in `PilotMain()` and store in global variable:

```
SysGetModuleDatabase(SysGetRefNum(), NULL, &gResDB);
```

To use for:

```
FrmGotoForm(gResDB, MainForm);
DmGetResource(gResDB, 'hair', 1000);
```



Minor Adjustments (continued)

No card numbers!

- Occurs in different parameter list positions
- Was also used in some data structures
- `MemNumCards()` always returns 1

Glue functions now included in API

Database creation

- `DmCreateDatabase()` creates an **extended** database
- `DmCreateDatabaseV50()` creates a **classic** database



Use Standard C String Library Functions

Not strictly necessary

- Some performance gains; more, um, standard

Not always correct

- See next slide

Can't always use Search & Replace method

- Remember the parameter list for `MemSet()`?

Can now use `malloc()`

- Instead of `MemPtrNew()` or `MemHandleNew()`

```
#include <string.h>
```



Palm OS® vs. Standard C Strings

Many `StrXXX()` functions just call standard function:

- `StrLen()` calls `strlen()`
- `StrCompareAscii()` calls `strcmp()`

But some respect/use localization and multi-byte characters:

- `StrCompare()` - localized string comparison
- `StrToLower/Upper()` - current local
- `StrChr()`, `StrStr()` - multibyte chars
- `StrNCopy()`, `StrNCat()` - won't truncate in middle of multibyte character



Changes With Side Effects

Different type name and/or different size

May cause loss of data or comparison errors

- If new size is different than old

New error type: `Err` is now `status_t`

- Is now 32-bit signed integer (`int32_t`)
- Positive values not used by Cobalt
- Might affect function return value type

Casting won't always fix, because…



Changes With Side Effects (continued)

`sysTicksPerSecond`

- Macro **`SysTicksPerSecond()`** returns 1000
- Always. Really. Everywhere.

`TimGetTicks()`

- Returns **`uint64_t`**

Many other 'size' types now use larger types

- **`uint16_t`** often changed to **`uint32_t`**, etc.
- Record index, preference entry size, …



Function Rewrite

`DmDatabaseInfo()`

- Used to have long list of attribute parameters
- Now uses `DmDatabaseInfoType` structure

`DmRecordInfo()`

- Gone, replaced with `DmGetRecordAttr()`, `DmGetRecordID()` and `DmGetRecordCategory()`

`WinSaveBits()`, `WinRestoreBits()`

- No longer available

Previously-deprecated functions are gone now

- E.g., `CategoryEditV20()`



Function Rewrite (continued)

`ListDrawDataFuncType`

- Used for custom list drawing
- Provides **`ListType`** pointer as parameter

New resource required for global Find

- Only the active app gets **`sysAppLaunchCmdFind`** by default
- Must add Application Launch Preferences **`('alpf' 0)`**
- Set the Find flag to TRUE
- Other flags: Reset, Time Change, Install and DB Change



Function Rewrite (continued)

NetLib is not in the Palm OS® Cobalt API

Use industry-standard Berkeley sockets API

Look at NetSocket.c in SDK 5 for help

Unsupported in Berkeley sockets API:

- AF_UNIX address family
- PF_UNIX protocol family
- `socketpair()`
- No UNIX-style asynchronous features (signals, options, flags)



Environmental Changes

App process is torn down when application quits

- Don't use `MemPtrSetOwner()`
- Don't use callback notifications
  - No guarantee that your code will be loaded, or loaded at the same location after an app switch
- Don't use embedded pointers in `SysNotifyParamType`

Background audio playback

- Don't use SndStream API callbacks to play after app quits
- Create background thread and play in background process

Attention Manager

- No callbacks! Only sublaunches allowed



Environmental Changes

Handle frmUpdateEvent for custom drawing

- Type `compat event update` in Terminal to expose

`winSwap, winInvert` deprecated

- If used for text or frames, nothing drawn on screen

Dynamic input area

- Finally, a single API
- Use window constraints to manage
- Once you use constraints, you cannot specify your window's size



Environmental Changes

Shared libraries

- All apps are now shared libraries, exporting `PilotMain()`
- Apps can export other functions
- Easier to write than before (no trap dispatch table)
- Can use ‘real’ global variables
- To use, link against library’s stub file

MathLib

- Has been ported to ARM
- But not in ROM in Palm OS Cobalt, version 6.0 release



Environmental Changes : ARM v. 68K

Integers and data structures

- Endianness: Palm OS Cobalt uses ARM in little endian mode
- Alignment of data (`struct` packing)
- For 68K apps, PACE handles this (when it can)

Impact

- Beaming/sending of binary data with big-endian devices
- Conduits (byte-swapping routines can handle)

Solution

- Develop wire format for data sharing (i.e., strings)
- Example: PIM apps use vCard, vCal



Environmental Changes: Application Process

Global variables are always available

- For all launch codes
- For shared libraries

Memory chunks no longer limited to < 64K

- Code segmentation no longer necessary

Although many API functions will be faster, there are exceptions, due largely to IPC

- Come to session 340 to hear all about IPC



Environmental Changes: PACE

No resource call chain

- Cannot use `DmGetResource()` or other calls to find system resources

Undocumented system resources may have different IDs or no longer exist

Some features are preserved for compatibility

- Standard Edit menu (ID 10000)
- Alert icon bitmaps

PNOs are supported



PIM Databases

Native PIM databases are now schema databases

When 68K app tries to access a 68K PIM database:

- PACE creates and opens a temporary 68K database
- Inits the AppInfoBlock
- Creates a cache of the schema database record info blocks

To read a record:

- PACE sends `sysAppLaunchCmdExportRecord` to PIM app
- PIM app sends vCard
- PACE converts vCard, stores in temp database

Temporary database deleted

- When 68K app closes database, or quits



Next Steps

Read *Porting Applications to Palm OS® Cobalt*

- Documents all header file changes
- Full description of PACE support for classic PIM databases

Come to the lab sessions

Start with a simple app first



Q & A



palm OS Cobalt

Contact: Doug MacMillan
Sr. Palm OS Platform Evangelist
Email: doug.macmillan@palmsource.com

Contest: Open Labs (Tues/Wed)
68K Compatibility & Protein Contests
Motorola sponsored Grand Prize!

MOTOROLA digital dna